MAXIMILIANO GOMEZ MONTERO

Manufacturing & Project Engineer | Smart Manufacturing

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Canadian Permanent Resident (no sponsorship required) • Toronto/GTA • Available January 2027

SUMMARY

Manufacturing engineer with hands-on experience taking industrial finishing systems from concept to start-up for EV and heavy-equipment plants. Engineering lead on 54 opportunities (USD 21M quoted). Brings data and automation to the shop floor: built digital tools that save ~340 hours a year and co-authored a peer-reviewed ML paper on machining quality prediction (Springer). M.S. in Manufacturing Engineering (GPA 4.0).

EXPERIENCE

Project Engineer – Dove Equipment / ICAFe, East Peoria, IL *Jun 2025 – Present*

Engineering Intern, Jan – May 2025

- Engineer industrial finishing systems (paint/powder booths, cure ovens, AMUs, wash systems, conveyors) for EV, heavy-equipment and building-products plants: process selection, airflow and thermal sizing, layouts and equipment specs.
- Engineering lead on 54 opportunities (USD 21M quoted; 10 of 23 decided projects won, USD 4.85M), taking awarded projects from concept to start-up with customers, sales, installers and OEM vendors (GFS, Rohner, Nordson, Clemco).
- Managed ~6 on-site installations: installers, on-site schedule, punch lists, burner and line start-ups and inspections, incl. a multi-booth EV paint line.
- Review drawings, submittals and quotes for scope and capacity gaps (e.g., hook load, ceiling height, electrical scope) before release.
- Standardized ERP project data with schedule-overrun alerts; built a historical database of 1,697 equipment items from 620 past projects.

Graduate Teaching Assistant – Bradley University, Peoria, IL *Jan 2024 – May 2025*

- Taught machining (milling, turning), injection molding and Excel/Power BI labs; supported curriculum design and research.

Logistics Analyst – ABC Distributions, Tucumán, Argentina *Apr 2023 – Dec 2023*

- Raised operational efficiency 15% through time studies and workforce optimization; digital tools lifted sales productivity 10% per worker.

PROJECTS

Lean Redesign Proposal (team of 5, not implemented) – Paint System Assembly – Bradley University *2025*

- Team of 5: mapped the assembly value stream and waste (muda); proposed a work cell, kanban and SOP (est. $16.5K/yr savings).

Digital tools built with AI – ProjectTracker and restaurant operations app *2026*

- Built a workflow app used daily to track 54 opportunities (Outlook, ERP and SharePoint integration, automated drawing QA); saves ~340 h/year.
- Built a desktop accounting/analytics app for a restaurant (POS and delivery-portal data import, OCR, payroll, cost per dish), in use since Oct 2026.

PUBLICATIONS

Chen, J., **Gomez Montero, M. A.**, & Yu, S. (2026). Development of an ANN-based online surface roughness prediction (ANN-OSRP) system in wire electrical discharge machining processes. *Neural Computing and Applications*, 38(4), 87. Springer.

- Neural network predicting wire-EDM surface roughness in real time from machine-controller signals; ~95.8% average accuracy.

Under review: vibration-based prediction of surface roughness and perpendicularity in CNC milling (Python ML models), *IJIDeM*.

EDUCATION

M.S. Manufacturing Engineering – Bradley University, Peoria, IL | GPA 4.0 *May 2025*

- Graduate Commencement Speaker (Spring 2025); Marjorie Reynolds Award for academic excellence.

Industrial Engineering Degree – National University of Tucumán, Argentina *Sep 2023*

- Final project (Industry 4.0): led a WMS/TMS digital transformation of order-to-delivery logistics workflows.

SKILLS

- **Manufacturing:** Finishing systems, process design, layout planning, lean/time studies, commissioning and start-up, CNC, wire-EDM, injection molding, PLCs, SOPs
- **Tools:** AutoCAD, FactoryCAD, Creo, ERP (Acumatica), SharePoint, Power BI, Excel
- **Data & Digital:** Python, SQL, ML (ANN, regression), Taguchi DOE, AI-assisted development (Claude Code), ERP/SharePoint data integration
- **Languages:** Spanish (native), English (CELPIP CLB 10), French (TEF Canada NCLC 7+)

HONORS & ACTIVITIES

- Assistant coach, Central Illinois United (girls U14–U17) and Illinois Central College women's soccer; US Soccer grassroots license.
- 1st Place, Latin American Innovation Rally 2021 (provincial level): tech-based solution to address domestic violence.